

Gear Optimization

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Abstract—This study investigates the optimization of spur gear geometry with fixed center distance, transmitted torque, and rotation speed. Both bending fatigue and contact fatigue are considered as governing failure modes. The objective is to determine optimal gear ratio and geometric parameters that maximize gear life while satisfying design constraints. A reference industrial spur gear system (center distance $a = 100$ mm, torque $T = 150$ N·m, speed $n = 1450$ rpm) is used to validate and demonstrate the optimization procedure.

Index Terms—Gear, Optimization, Grid Search, Bending stress, Contact stress

I. INTRODUCTION

Gear systems are widely used in power transmission applications. Their performance is governed by two primary failure modes: tooth root bending fatigue and surface contact fatigue. This study aims to evaluate whether a given spur gear system is optimally designed and to explore improvements through parameter optimization. Standard AGMA and Lewis formulations are used for stress computation, and S-N based fatigue life equations are employed to quantify gear life.

II. REFERENCE MODEL

To ground the optimization in a concrete case, the following reference parameters are adopted (representative of a light industrial speed reducer):

TABLE I: Reference model parameters used in the simulation

Parameter	Symbol	Value
Center distance	a	100 mm
Input torque	T	150 N·m
Input speed	n	1450 rpm
Pressure angle	ϕ	20°
Material	-	Steel (Grade 1)
Young's modulus	E	206 GPa
Poisson's ratio	ν	0.3
Allowable bending stress	σ_b	220 MPa
Allowable contact stress	σ_c	900 MPa
Overload factor	K_o	1.25
Reliability factor	K_r	1.0
Gear quality number	Q_v	6

III. PROBLEM FORMULATION

The objective is to maximize the fatigue life of a meshing spur gear pair by optimizing the design parameters subject to geometric and stress constraints.

A. Known Parameters

- Center distance (a)
- Transmitted torque (T)
- Rotation speed (n)
- Material properties (E , ν , $\sigma_{b,allow}$, $\sigma_{c,allow}$)

B. Design Parameters

- Gear ratio ($i = Z_1/Z_2$)
- Module (m)
- Face width (b)

All other quantities — tooth counts, pitch radii, velocities — are derived from these three.

C. Derived Quantities

The number of teeth on the pinion and gear are:

$$Z_1 = \frac{2a}{m(1+i)}, \quad Z_2 = i \cdot Z_1 \quad (1)$$

The pitch radii are:

$$r_{p1} = \frac{a}{1+i}, \quad r_{p2} = \frac{ai}{1+i} \quad (2)$$

The pitch line velocity is:

$$V = \frac{\pi r_{p1} n}{30} \quad [\text{m/s}] \quad (3)$$

The tangential load is:

$$F_t = \frac{T}{r_{p1}} \quad (4)$$

D. Constraints

$$a = \frac{m(Z_1 + Z_2)}{2} \quad (5)$$

$$i = \frac{Z_2}{Z_1} \quad (6)$$

$$\varepsilon \geq 1.2 \quad (7)$$

$$\sigma_b \leq \sigma_{b,allow} \quad (8)$$

$$\sigma_c \leq \sigma_{c,allow} \quad (9)$$

$$8m \leq b \leq 12m \quad (10)$$

$$Z_1 \geq 17 \quad (11)$$

Constraint (11) prevents undercutting of the pinion teeth, which is a standard requirement for 20° pressure angle spur gears.

1) *Contact Ratio Formula*: The contact ratio ε in Constraint (7) is computed as:

$$\varepsilon = \frac{\sqrt{r_{a1}^2 - r_{b1}^2} + \sqrt{r_{a2}^2 - r_{b2}^2} - a \sin \phi}{\pi m \cos \phi} \quad (12)$$

where the addendum radii and base circle radii are:

$$r_a = r_p + m, \quad r_b = r_p \cos \phi \quad (13)$$

E. Objective Function

The gear system life is governed by whichever failure mode occurs first:

$$\text{Life} = \min(N_b, N_c) \quad (14)$$

where N_b and N_c are the number of cycles to failure under bending and contact fatigue respectively (defined in Section ??). The optimization goal is:

$$\max_{i, m, b} \min(N_b, N_c) \quad \text{subject to constraints (5)–(11)} \quad (15)$$

IV. STRESS MODELS

A. Modified Lewis Bending Stress

Accounting for dynamic and application loading effects, the modified Lewis equation is:

$$\sigma_b = \frac{F_t \cdot K_o \cdot K_s \cdot K_H \cdot K_v}{b \cdot m \cdot J} \quad (16)$$

where each factor is defined as follows:

- F_t — Tangential load (N), see Eq. (20) below.
- K_o — **Overload factor**: accounts for external dynamic loads beyond nominal (e.g. shock from the prime mover). Typically 1.0–1.75 depending on the character of the power source and driven machine.
- K_v — **Dynamic factor**: captures internal dynamics from tooth-mesh impact. Using the AGMA standard,

$$K_v = \left(\frac{A + \sqrt{V}}{A} \right)^2 \quad (17)$$

where V is pitch line velocity in m/s, and A is an AGMA-defined coefficient derived from the gear quality number Q_v as:

$$A = 50 + 56 \left(1 - \frac{Q_v}{12} \right) \quad (18)$$

- K_s — **Size factor**: accounts for non-uniform material properties in larger teeth. Taken as 1.0 for module $m < 5$ mm; otherwise determined from AGMA tables.
- K_H — **Load distribution factor**: accounts for non-uniform load across the face width due to misalignment and deflection. For accurate mounting with face widths up to 125 mm, $K_H = 1.6$ is adopted (AGMA recommendation).
- b — Face width (mm).
- m — Module (mm).
- J — **Geometry factor**: accounts for tooth form and stress concentration at the root fillet. For a 20° pressure angle,

J is approximated (as a curve fit of the tabulated values given by the AGMA) as:

$$J \approx 0.32 \ln(Z) - 0.16 \quad (Z \geq 17, \phi = 20^\circ) \quad (19)$$

Since Z is a derived quantity (a function of m and i), J varies across the design space and must be recomputed at each iteration.

The tangential load is:

$$F_t = \frac{T}{r_{p1}} \quad (20)$$

B. Contact Stress (AGMA Model)

The surface contact stress is evaluated using the AGMA formulation for pitting resistance:

$$\sigma_c = Z_E \sqrt{\frac{F_t \cdot K_o \cdot K_v \cdot K_s \cdot K_H}{b \cdot d_p \cdot I}} \quad (21)$$

where:

- F_t — Tangential load (N)
- K_o — Overload factor
- K_v — Dynamic factor
- K_s — Size factor
- K_H — Load distribution factor
- b — Face width (mm)
- d_p — Pitch diameter of the pinion (mm)
- I — Geometry factor for pitting resistance
- Z_E — Elastic coefficient

The pitch diameter of the pinion is given by:

$$d_p = 2r_{p1} \quad (22)$$

The elastic coefficient Z_E is defined as:

$$Z_E = \sqrt{\frac{1}{\pi \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \right)}} \quad (23)$$

For identical materials ($E_1 = E_2 = E$, $\nu_1 = \nu_2 = \nu$):

$$Z_E = \sqrt{\frac{E}{2\pi(1-\nu^2)}} \quad (24)$$

The geometry factor I for spur gears is approximated as:

$$I = \frac{\cos \phi \cdot \sin \phi}{2} \cdot \frac{i}{1+i} \quad (25)$$

This formulation accounts for material properties, gear geometry, and load distribution effects, and is consistent with AGMA standards for evaluating surface fatigue (pitting) failure.

C. Bending Fatigue Life

Using a power-law S-N model (Basquin's equation):

$$N_b = \left(\frac{\sigma_{b, \text{allow}}}{\sigma_b} \right)^{m_{\text{SN}}} \cdot N_0 \quad (26)$$

where $m_{\text{SN}} = 3.5$ is the S-N slope exponent for steel gear teeth (AGMA Grade 1), and $N_0 = 10^7$ cycles is the reference endurance cycle count.

D. Contact Fatigue Life

For surface fatigue (pitting):

$$N_c = \left(\frac{\sigma_{c,allow}}{\sigma_c} \right)^{m_c} \cdot N_0 \quad (27)$$

where $m_c = 6$ is the S-N exponent for contact fatigue in steel (Hertz-based pitting model), and $N_0 = 10^7$ cycles.

These exponents are standard values from AGMA 2101-D04 and Shigley's *Mechanical Engineering Design*.

V. OPTIMIZATION PROCEDURE

The problem is solved using a grid search over the discrete variable m (standard module values) and continuous variables i and b . For each feasible combination:

- 1) Compute $Z_1, Z_2, r_{p1}, r_{p2}, V, F_t, F_n$.
- 2) Check all constraints (Eqs. (7)–(11)).
- 3) Compute σ_b (Eq. (16)) and σ_c (Eq. (25)).
- 4) Compute N_b (Eq. (26)) and N_c (Eq. (27)).
- 5) Record Life = $\min(N_b, N_c)$.

The design with maximum Life among all feasible points is selected as the optimum.

VI. RESULTS AND DISCUSSION

The optimization was run over 8 standard module values, and 5 face width fractions, yielding approximately 8000 total design evaluations.

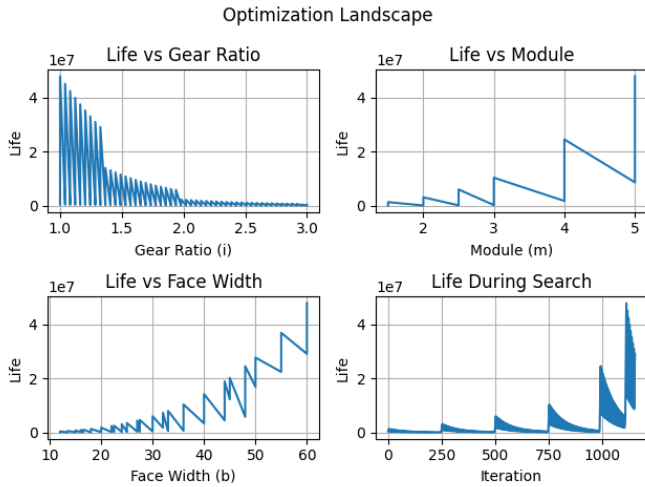


Fig. 1: Life vs Gear Ratio, Module, Face Width, iterations

Table II and Table III summarises the baseline and optimal designs. The transmitted torque ($T = 150$ N·m) and input speed ($n = 1450$ rpm) are identical in both cases; only the geometric parameters differ.

TABLE II: Geometric comparison of baseline and optimal designs

Parameter	Baseline	Optimal
Gear ratio i	2.000	1.000
Module m (mm)	2.00	5.00
Face width b (mm)	20.00	60.00
Teeth Z_1	33	20
Teeth Z_2	67	20
Contact ratio ε	1.739	1.557
Pitch velocity V (m/s)	5.06	7.59

The following life was obtained

TABLE III: Stress and fatigue life comparison

Parameter	Baseline	Optimal
Dynamic factor K_v	1.059	1.072
Geometry factor J	0.962	0.799
Bending stress σ_b (MPa)	247.5	26.84
Contact stress σ_c (MPa)	1550	693.3
N_b (cycles)	6.62×10^6	1.58×10^{10}
N_c (cycles)	3.83×10^5	4.79×10^7
Life ($\min(N_b, N_c)$)	3.83×10^5	4.79×10^7

The baseline design is infeasible: its contact stress of 1550 MPa significantly exceeds the allowable limit of 900 MPa, indicating that the gear pair would fail by surface pitting very early in service. The predicted life of 3.83×10^5 cycles reflects this severe overstressing. The bending stress (247.5 MPa) also exceeds the allowable value, though contact fatigue remains the dominant failure mode.

The optimized design brings both stresses within acceptable limits. Contact stress is reduced by approximately 55% to 693 MPa, while bending stress is reduced by nearly 90% to 26.8 MPa. The resulting life of 4.79×10^7 cycles represents an improvement of approximately $125\times$ relative to the baseline. This improvement is primarily driven by the increase in module from 2 mm to 5 mm and the corresponding increase in face width, which together distribute the load over a larger contact area and reduce both bending and contact stresses.

It is worth noting that the mathematical optimum occurs at $i = 1$ (equal gears), since a lower gear ratio increases the pinion pitch radius and thereby reduces the tangential load $F_t = T/r_{p1}$. In practice, however, the gear ratio is typically fixed by the application's required speed reduction rather than treated as a free variable. The optimization should therefore be repeated with i fixed at the desired value, optimizing only over m and b .

VII. CONCLUSION

This study optimized the geometry of a spur gear pair for maximum fatigue life under fixed operating conditions ($T = 150$ N·m, $n = 1450$ rpm, $a = 100$ mm). The three design variables — gear ratio i , module m , and face width b — were optimized subject to stress, contact ratio, and geometric feasibility constraints.

The baseline design ($i = 2$, $m = 2$ mm, $b = 20$ mm) was found to be infeasible, with both contact and bending stresses exceeding their allowable limits. The contact stress of 1550 MPa is significantly higher than the allowable value, resulting

in a predicted life of only 3.83×10^5 cycles, with contact fatigue remaining the governing failure mode.

The optimized design ($i = 1$, $m = 5$ mm, $b = 60$ mm) satisfies all constraints and achieves a gear life of 4.79×10^7 cycles — an improvement of approximately $125\times$ relative to the baseline — with contact fatigue remaining the limiting failure mode in both cases.

Three design conclusions follow. First, module is the most influential parameter: increasing m from 2 mm to 5 mm accounts for the majority of the stress reduction by enlarging the tooth and spreading the contact load. Second, face width should be increased toward the upper feasibility limit ($b = 12m$), as it directly reduces contact stress, though practical considerations such as alignment and manufacturing must be considered. Third, since contact fatigue consistently governs life across the design space, further improvements beyond geometry — such as surface hardening or use of higher-grade materials — should specifically target reduction of σ_c .

VIII. LIMITATIONS

While the present study provides useful insights into the influence of geometric parameters on gear fatigue life, several simplifying assumptions and modeling limitations must be acknowledged.

First, the face width b is only constrained within a proportional range ($8m \leq b \leq 12m$), without considering practical limits such as shaft deflection, misalignment sensitivity, or packaging constraints. In real applications, excessively large face widths can lead to non-uniform load distribution, increased manufacturing complexity, and higher sensitivity to mounting errors. As a result, the optimal value $b = 12m$ obtained here represents an upper bound rather than a universally practical design choice.

Second, the gear ratio i is treated as a free design variable. This leads to a mathematical optimum at $i = 1$, which minimizes the tangential load by maximizing the pinion radius. However, in most practical systems, the gear ratio is dictated by functional requirements such as desired speed reduction or torque multiplication. The current formulation therefore represents a theoretical optimum rather than a directly applicable design solution, and the optimization should be repeated with i fixed for specific applications.

Third, the analysis assumes ideal material behavior and constant material properties. Effects such as thermal fluctuations, temperature-dependent material degradation, and lubrication conditions are not considered. In practice, temperature variations can significantly influence contact stress, wear rates, and fatigue life, particularly in high-speed or heavily loaded gear systems.

Fourth, the stress models are based on AGMA formulations with simplified geometry factors. While these provide reliable engineering estimates, they do not capture detailed stress distributions, local stress concentrations, or micro-geometry effects such as profile modifications and surface roughness. More accurate analysis would require finite element modeling or experimental validation.

Finally, the optimization is performed using a grid search approach over a discretized design space. While this ensures coverage of the feasible region, it may miss finer variations between grid points and does not guarantee identification of a true continuous optimum. However, the smooth trends observed in the results suggest that the identified solution is representative of the global optimum within the chosen resolution.

These limitations indicate that the present results should be interpreted as a preliminary design study. Further refinement incorporating additional constraints, detailed modeling, and application-specific requirements would be necessary for final engineering implementation.

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